

4.1 Chapter Outline and Structural Logic

This chapter presents the empirical and simulation findings derived from the methodology described in Chapter 3. The presentation follows a hypothesis-driven structure, where each research sub-question is addressed sequentially. Quantitative results are reported with statistical precision, while qualitative observations (e.g., convergence behaviour, transient response) are visualised using high-resolution figures. The discussion interprets these findings, compares them against baselines and prior literature, and articulates the study's tangible contributions.

4.2 Presentation of Results

4.2.1 Experimental/Simulation Setup and Data Provenance

- Restate key parameters: sampling frequency, solver type, time step, solver tolerance, dataset split, hyperparameters.
- Specify the execution environment: hardware, software, libraries.
- Confirm reproducibility: "All simulations were executed in [software/hardware specification]."

4.2.2 Baseline Definitions and Comparative Algorithms

- Define reference models (e.g., PID, MPC, classical SVM, plain CNN, rule-based protection schemes).
- Specify performance metrics per specialization:
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Specialization	Recommended Metrics
Control	Settling time, overshoot, steady-state error, RMSE, ITAE
Electronics	THD, power efficiency, ripple voltage, response time, thermal sensitivity
Software Eng.	F1-score, classification accuracy, inference latency, memory footprint, generalization
Power	THD, power factor, frequency stability, voltage sag, fault recovery time

4.2.3 Quantitative Results (Tables and Figures)

- **Table 4.1:** Comparative performance (mean $\pm$ std) across three independent runs.
- **Figure 4.2:** Convergence curves of the proposed algorithm vs. baseline.
- **Figure 4.3:** Transient response (Control/Power).
- **Figure 4.4:** Confusion matrix (Electronics/Software).
- **Guideline:** Every figure must have a self-contained caption: "Figure 4.2 – Convergence of loss function for the proposed DDPG-based AVR controller over 500 episodes."

4.2.4 Statistical Significance and Robustness Checks (Mandatory for Excellence)

- Report p-values (t-test or Wilcoxon) for pairwise comparisons.
- Perform sensitivity analysis: "The proposed algorithm maintained <5% RMSE degradation under $\pm 20\%$ parameter variation."

4.3 Discussion of Findings

4.3.1 Interpretation in Light of the Proposed Framework

- Explain *why* observed improvements occur. Link results to model architecture or algorithmic innovation.
- Example (Control): "The 32% reduction in settling time is attributed to the adaptive gain scheduling enabled by the deep Q-network, which anticipates non-linear friction effects not modelled in the classical PID."

4.3.2 Comparison with Prior Work (Mandatory for Excellence)

- Construct a comparative table against 3–5 key papers from Chapter 2.
- **Table 4.2:** Qualitative and quantitative comparison with existing studies.
- Use explicit language: "Unlike the approach in [15], which requires full state observation, our method operates with partial observability by integrating a recurrent state estimator."

4.3.3 Addressing Discrepancies and Unexpected Results (Mandatory for Excellence)

- Do not hide anomalies. Explain them honestly.
- Distinguish between method limitations and validation limitations.
- Example (Electronics): "The slight increase (1.2%) in THD at low load is due to the switching ripple of the GaN device, which was not penalised in the cost function – a limitation we acknowledge and address in Section 5.3."

4.3.4 Articulation of the Actual Contribution (Mandatory for Excellence)

- **Separate paragraph titled "Actual Contributions of This Work"**
- Format as numbered, verifiable points.
- Example (Power/Control): "The actual contributions validated empirically are: (i) a hybrid physics-informed neural network for power system state estimation that reduces error by 41% compared to weighted least squares; (ii) the first experimental demonstration of a transformer-based fault diagnosis on an FPGA with <1 ms latency; (iii) an open-source implementation of the proposed software-defined protection logic."

4.4 Summary of Chapter Four

- Bullet-point summary of the 3–6 most important findings.
- Bridge sentence to Chapter 5: "Having established the efficacy of the proposed methods, the following chapter concludes the thesis and outlines directions for future research."

Chapter 5: Conclusion and Future Directions

5.1 Chapter Overview

This chapter synthesises the principal findings of the thesis, restates the original contribution without introducing new data, acknowledges limitations, and provides actionable, technically precise recommendations for subsequent research and engineering practice.

5.2 Summary of Achievements (Against Objectives)

- Revisit the three to four objectives from Chapter 1.
- Use a **numbered list** or **table** mapping objectives → results.

Example (Power):

1. *Objective 1*: Design a decentralised control for microgrids → Achieved (settling time improved by 27%).
2. *Objective 2*: Reduce harmonic distortion under unbalanced load → Achieved (THD <3% for all tested cases).

5.3 Limitations and Critical Reflection (Mandatory for Excellence)

- Explicitly state what the proposed system does *not* do well.
- Distinguish between:
 - **Methodological limitations**: Simplified assumptions (e.g., ideal communication links, linearity)
 - **Validation limitations**: Data or experimental constraints (e.g., imbalanced dataset, limited test scenarios)

Example: "The proposed control law assumes ideal communication links; performance under packet loss was not evaluated. Furthermore, the dataset for fault classification, while publicly available, is imbalanced with respect to rare fault types."

5.4 Recommendations for Future Research (Actionable Directions)

5.4.1 Short-term Extensions (1–2 years)

- **Algorithmic**: "Implement the same controller with event-triggered learning to reduce computational load by an estimated 60%."
- **Hardware**: "Port the neural network to a lower-cost STM32 microcontroller for edge deployment."
- **Validation**: "Collect experimental data from a physical 5 kW prototype under real temperature cycling."

5.4.2 Long-term Research Avenues (3–5 years)

- **Theoretical**: "Extend the stability proof to non-stationary environments using Lyapunov drift analysis."

- **Interdisciplinary:** "Integrate the fault diagnosis module with digital twin frameworks for predictive maintenance."
- **Open challenges:** "Address the explainability gap for safety-critical approval (e.g., IEC 61508 certification)."

5.5 Final Concluding Remarks

One concise paragraph (maximum 150 words) that echoes the opening of the thesis but now with the weight of evidence.

Example (applicable to all four specializations): "This thesis demonstrated that merging model-based control with deep reinforcement learning yields superior performance in non-linear power electronic systems. By providing validated open-source code and a reproducible benchmark, this work lays a foundation for certifiable AI in energy grids. The findings confirm that hybrid intelligence – not purely model-free or purely model-based – represents the most promising path for next-generation electrical engineering systems."